

WCMS UNIT-3

2M:

1. Define dynamic website. Write example.

=A dynamic website is a type of website that generates and delivers content to users in real-time, allowing for a more interactive and personalized user experience. Unlike static websites, where the content remains fixed and unchanged unless manually updated by a developer, dynamic websites can adapt and respond to user inputs, database changes, or other real-time events.

2. What is content generation?

=Dynamic websites generate content on the server side in response to user requests or interactions. Content is often retrieved from databases or external sources and can vary based on user preferences, actions, or the current state of the website.

3. Define static website. Write example.

=A static website is a type of website that displays fixed content to users and remains the same unless manually updated by a developer. The content on static websites doesn't change dynamically based on user interactions or real-time events. Each page of a static website is prebuilt and stored as HTML files, and when a user requests a page, they receive the same content that was initially created by the developer.

4. Mention any two characteristics of dynamic website.

- Content Generation
- Interactivity
- Database Integration
- User Authentication and Accounts

5. Mention any two characteristics of static website.

- Fixed Content
- Simplicity
- Ease of Hosting
- Page Loading Speed

6. Differentiate static website and dynamic website.

=Refer long answers.

7. Give two examples each for static and dynamic websites.

=Static websites:

- Brochure websites, where the goal is to provide information about a business or organization.
- Personal portfolios or resume websites.
- Landing pages for marketing campaigns with fixed content.

Dynamic websites:

- Social media platforms, e-commerce sites, news websites, and web applications with user accounts.

8. Expand AJAX. What is the use of it?

=**Asynchronous JavaScript and XML**

- Improved User Experience:AJAX can load new data into a webpage without refreshing the whole page. For example, when you scroll through your social media feed and new posts appear automatically.
- Reduced Server Load:Only the necessary data is sent to and from the server, which reduces the amount of data transferred and the load on the server.

9. What is Wordpress?

=WordPress is a website creation platform that allows users to customize most of what they want to showcase on their page. It's a dynamic website because it automatically refreshes the content in real-time, not only on your end but from the visitors' point of view.

10. What is Twitter?

=Twitter, much like Facebook, alters the content you see based on different factors like your search queries and public and private lists. In addition, your engagement, such as retweets and likes, also prompts Twitter to generate content related to your past actions.

11. Give four examples for server-side language.

=PHP, Python, Ruby, and Node.js.

12. Mention any two front-end development language and framework.

=e HTML, CSS, and JavaScript.

13. What is CSS? List types of CSS.

=Cascading Style Sheets (CSS) is a stylesheet language used to describe the presentation of a document written in HTML or XML (including XML dialects such as SVG, MathML or XHTML). CSS describes how elements should be rendered on screen, on paper, in speech, or on other media.

14. What are the 3 types of CSS?

- Inline CSS
- Internal or Embedded CSS
- External CSS

15. What is the use of Type selector and Attribute selectors?

=**Type selector:**

Basic Styling: Applying styles to all elements of a certain type. For example, setting the font size for all heading tags.

Consistency: Ensuring a consistent look and feel for certain HTML elements across a webpage or website.

Base Styles: Providing default styles that can be overridden by more specific selectors later

Attribute selector:

Form Styling: Applying specific styles to form elements based on their type, like styling text inputs differently from submit buttons.

Enhanced Selectivity: Targeting elements more precisely without needing additional classes or IDs. For example, applying styles to links that open in a new tab.

Dynamic Content: Styling elements based on dynamically generated attributes, such as highlighting active elements or those that meet specific criteria.

16. What is the use of class selector and child selectors?

=Class selector:

Reusable Styles: Applying the same style to multiple elements, regardless of their type. For example, adding a consistent style to elements that need to stand out.

Component-Based Design: Styling specific components of a web page that share the same class. For example, styling all buttons with the class `.btn-primary`.

Modular CSS: Creating styles that can be easily reused and combined with other classes. For example, using utility classes like `.text-center` or `.mt-4` for margin top spacing.

Child selector:

Specific Hierarchical Styling: Applying styles to elements only if they are direct children of a specific parent. This ensures that nested elements or elements further down the hierarchy are not affected.

Enhanced Precision: Targeting specific elements in complex layouts where general selectors would not suffice. For example, styling only the direct child list items of a specific `<ul>` without affecting nested lists.

Avoiding Over-Targeting: Ensuring that styles are not inadvertently applied to deeper nested elements, which might cause unintended design changes.

17. Define a blog? Give examples.

=A blog is a type of website or a section of a website where individuals, businesses, or organizations regularly publish content in a conversational and informal style. Blogs serve various purposes, and their use can depend on the goals and intentions of the blogger.

- Personal Blogs
- Professional Blogs
- Corporate Blogs
- Travel Blogs:
- News Blogs

18. Write any two purposes of blogs.

- Content Sharing
- Personal Expression

- Business Marketing
- Search Engine Optimization (SEO)

19. What is content sharing and Search Engine Optimization (SEO)?

=Content Sharing:

Blogs are platforms for sharing information, ideas, and knowledge on specific topics.

Bloggers often publish articles, guides, and tutorials to educate and inform their audience.

Search Engine Optimization (SEO):

Regularly updated blogs can improve a website's SEO. Search engines favor fresh and relevant content, and blogs provide an opportunity to incorporate keywords, driving organic traffic to the site.

20. How community sharing is facilitated through blogs?

=Comment Sections:Most blogs have comment sections where readers can share their thoughts, ask questions, and provide feedback on blog posts. This encourages discussions and allows readers to interact with the author and each other.

Social Media Integration:Blogs often include social media sharing buttons, making it easy for readers to share content on platforms like Facebook, Twitter, Instagram, and LinkedIn. This helps spread the blog's content to a wider audience and attracts new readers.

Interactive Content: Interactive elements like quizzes or challenges can engage readers in a fun way, encouraging them to share their results and experiences with the community.

Events and Webinars:Hosting webinars, live streams, or virtual meetups allows the community to interact in real-time, ask questions, and engage more deeply with the blog's content.

21. What is monetization?

=Monetization of blogs refers to the process by which bloggers generate revenue from their online content. This involves various strategies that turn their blog traffic and reader engagement into income. Common methods include displaying ads, either through ad networks like Google AdSense or by selling ad space directly to companies. Bloggers can also earn money through sponsored content, where companies pay them to write posts promoting their products or services.

22. Write the steps to customize blogs.

=Select a Theme:

Choose a visually appealing and responsive theme for your blog. Most platforms offer a variety of free and premium themes.

Customize Design Elements:

Personalize your blog by customizing colors, fonts, and layout. Make sure your design aligns with your blog's theme and target audience.

23. Give two examples each of blog sites for

i) Technology and Business

- TechCrunch
- Mashable

ii) Personal development and lifestyle.

- Tiny Buddha
- Zen Habits

24. Give two examples each of blog sites for –

i) Parenting & Family

- Scary Mommy
- The Bump

ii) Travel.

- Nomadic Matt
- The Points Guy

LONG ANSWER QUESTIONS:

1. Explain the characteristics of dynamic websites. (4/6)

1. Content Generation:

- Dynamic websites generate content on the server side in response to user requests or interactions.
- Content is often retrieved from databases or external sources and can vary based on user preferences, actions, or the current state of the website.

2. Interactivity:

- Dynamic websites are highly interactive, allowing users to engage with various features, submit forms, and experience real-time updates.
- Interactivity is achieved through server-side scripting languages (e.g., PHP, Python, Ruby) and client-side scripting languages (e.g., JavaScript).

3. Database Integration:

- Dynamic websites frequently use databases to store and manage data.
- User-generated content, account information, and other dynamic elements are often stored and retrieved from databases to provide a personalized experience.

4. User Authentication and Accounts:

- Dynamic websites often include user authentication systems, enabling users to create accounts, log in, and access personalized content.
- User sessions and account information are managed on the server to ensure security and privacy.

5. E-commerce Functionality:

- Many dynamic websites are designed for e-commerce, allowing users to browse products, add items to a shopping cart, and complete transactions.
- Shopping carts, order processing, and inventory management are handled dynamically

6. Content Management Systems (CMS):

- Dynamic websites often employ Content Management Systems to facilitate easy content updates.
- CMS allows website administrators to add, edit, or remove content without the need for extensive coding or technical knowledge.

7. Real-time Updates:

- Information on dynamic websites can update in real-time, providing users with the latest data, news, or changes.
- This real-time functionality enhances the user experience and keeps the content relevant

8. Scalability:

- Dynamic websites can handle a large amount of data and traffic, making them suitable for scalable and complex web applications.

2. Explain the characteristics of static websites. (4/6)

1. Fixed Content:

- Content on static websites is constant and does not change automatically.
- Any modifications or updates to the content require manual editing of the HTML files.

2. Simplicity:

- Static websites are typically simpler to create and maintain, making them suitable for smaller websites or those with minimal content.

3. Ease of Hosting:

- Static websites can be hosted on simple web servers without the need for complex server-side technologies.
- They are often hosted on platforms that specialize in serving static content.

4. Page Loading Speed:

- Static websites generally load faster than dynamic websites since the content is pre-built and doesn't require server-side processing for each user request.

5. Security:

- Due to their simplicity, static websites can be considered more secure as there are fewer vulnerabilities associated with server-side scripting or database interactions.

6. Cost-Effectiveness:

- Static websites are cost-effective to host and maintain because they don't require server-side processing or database management.

7. Limited Interactivity:

- Interactivity on static websites is limited to hyperlinks and basic HTML forms.
- Features like user accounts, dynamic forms, or personalized content are not inherent in static websites.

8. Examples:

- Brochure websites, where the goal is to provide information about a business or organization.
- Personal portfolios or resume websites.
- Landing pages for marketing campaigns with fixed content.

3. Explain any four/six differences between static and dynamic websites. (4/6)

Static Website	Dynamic Website
The content remains fixed and doesn't change unless manually updated by a developer	Content can change dynamically based on user interactions, database updates, or other real-time events.
Built using HTML and CSS. It's simple and easy to create but lacks interactive features.	Involves more complex coding with server-side scripting languages like PHP, Python, or Ruby, enabling dynamic content generation.
Generally faster since all content is pre-built and doesn't require server processing for each user request.	May have slower loading times as content is generated on the server based on user requests.
Manual updates are required for content changes. Each page needs to be individually modified.	Content updates can be automated, and changes are reflected across the site without the need to modify each page individually
Typically, easier to scale as it involves serving pre-built files. Suitable for small to medium-sized websites.	May require more server resources as the server has to process requests and generate content dynamically. Suitable for larger and more complex websites.
Limited interactivity. Interaction is mostly limited to hyperlinks.	Highly interactive. Users can engage with forms, submit data, and experience real-time updates.
Brochure websites, portfolios, personal blogs with fixed content.	Social media platforms, e-commerce sites, news websites, and web applications with user accounts.

4. Write a note on any THREE dynamic websites. (6)

1. Facebook generates personalized content based on the user's actions. The things you see on the website change depending on the accounts you follow, the content you're engaging with, or the people you have on your friends list.

2. Twitter, much like Facebook, alters the content you see based on different factors like your search queries and public and private lists. In addition, your engagement, such as retweets and likes, also prompts Twitter to generate content related to your past actions.

3. YouTube utilizes an algorithm that alters your feed based on the accounts you subscribe to, the videos you watch, and other related content that aligns with your viewing habits.

Google is the world's most popular search engine and one of the best examples of a dynamic website. The search engine indexes billions of websites, both new and refreshed, utilizing its algorithm to select, rank, and generate the best results for user-based queries.

4. WordPress is a website creation platform that allows users to customize most of what they want to showcase on their page. It's a dynamic website because it automatically refreshes the content in real-time, not only on your end but from the visitors' point of view.

5. Illustrate the steps for creating dynamic web content. (6)

=Creating dynamic web content involves using technologies and techniques that allow your website to display and update content in real-time based on user interactions, database changes, or external data sources. Here's a general guide on how to create dynamic web content:

1. Choose a Server-Side Language:

Select a server-side scripting language such as PHP, Python, Ruby, or Node.js. This is where you'll handle server-side logic.

2. Set Up a Server:

- Choose a web server (e.g., Apache, Nginx) to host your dynamic web application.
- Configure your server to handle the chosen server-side language.

3. Database Integration:

- Choose a database system (e.g., MySQL, PostgreSQL, MongoDB) to store and retrieve dynamic content.
- Establish a connection between your server-side language and the database.

4. Server-Side Programming:

- Write server-side scripts to generate dynamic content based on user requests.
- Use server-side logic to query the database, process data, and generate dynamic HTML or other markup.

5. Front-End Development:

- Use HTML, CSS, and JavaScript to create the user interface.
- Employ a front-end framework like React, Angular, or Vue.js for a more interactive experience.

6. AJAX (Asynchronous JavaScript and XML):

- Implement AJAX to enable asynchronous data exchange between the browser and the server.
- Use JavaScript to update specific parts of the page without requiring a full page reload.

7. Web APIs:

- Create or integrate with web APIs to fetch data from external sources dynamically.
- APIs allow you to access and manipulate data from services like social media platforms, weather APIs, etc

8. Content Management Systems (CMS):

- Consider using a CMS (e.g., WordPress, Drupal) that provides a user-friendly interface for managing dynamic content.
- Customize your CMS to integrate with your dynamic content needs.

9. User Authentication and Authorization:

- Implement user authentication to allow personalized content for logged-in users.
- Control access to different parts of your dynamic content based on user roles.

10. Caching:

- Implement caching mechanisms to enhance performance by storing frequently accessed dynamic content.
- Consider using caching systems like Redis or Varnish.

11. Testing:

Regularly test your dynamic web content across different browsers and devices to ensure compatibility and responsiveness.

12. Security:

Implement security best practices, including input validation, protection against SQL injection, and secure data transmission (HTTPS).

13. Scalability:

Design your dynamic web application to be scalable, allowing it to handle increased traffic and data.

14. Monitoring and Analytics:

Implement monitoring tools and analytics to track user interactions, performance, and potential issues.

15. Continuous Improvement:

Regularly update and improve your dynamic content based on user feedback and changing requirements.

6. Explain selectors in CSS. (4)

=Selectors in CSS (Cascading Style Sheets) are used to target HTML elements for styling. They define which part of the HTML a set of CSS rules should apply to.

Here are the main types of CSS selectors:

1. Universal Selector:

* - Selects all elements on a page.

2. Type Selector:

h1, p, div - Selects elements based on their tag name.

3. Class Selector:

.classname - Selects elements with a specific class attribute.

4. ID Selector:

#id - Selects a single element with a specific ID attribute.

5. Attribute Selector:

[attribute=value] - Selects elements based on the presence or value of a specific attribute.

6. Descendant Selector:

ancestor descendant - Selects all descendants of a given ancestor.

7. Child Selector:

parent > child - Selects all direct children of a parent element.

8. Adjacent Sibling Selector:

element + element - Selects an element that is immediately preceded by a specified sibling element.

7. Give detailed steps to create website using CSS. (6)

1. HTML Structure:

- Create an HTML file with the basic structure of your webpage.
- Include elements like <header>, <nav>, <section>, and <footer>.

2. CSS Styling:

- Create a separate CSS file and link it to your HTML file.
- Style your webpage elements using selectors like body, header, nav, etc.

3. Basic Styling:

Apply basic styling to elements, such as setting font styles, background colors, and spacing.

4. Content and Links:

- Replace placeholder content in your HTML file with your actual text.
- Create navigation links using <a> tags.

5. Preview:

- Open your HTML file in a web browser to preview your website.
- Make adjustments to HTML and CSS as needed.

6. Optional Enhancements:

- Add JavaScript for interactivity (optional).
- Implement responsive design for different screen sizes.
- Consider SEO optimizations, such as meta tags.

8. Write a note on the purposes for which blogs are used. (4/6)

1. Content Sharing:

Blogs are platforms for sharing information, ideas, and knowledge on specific topics.

Bloggers often publish articles, guides, and tutorials to educate and inform their audience.

2. Personal Expression:

Many individuals use blogs as a means of personal expression. They share their thoughts, experiences, and perspectives on a wide range of topics, from travel and lifestyle to personal development and creative pursuits.

3. Business Marketing:

Blogs are valuable tools for businesses to promote their products or services. They can publish content that highlights their expertise, industry insights, and solutions, building trust with potential customers.

4. Search Engine Optimization (SEO):

Regularly updated blogs can improve a website's SEO. Search engines favor fresh and relevant content, and blogs provide an opportunity to incorporate keywords, driving organic traffic to the site.

5. Community Building:

Blogs facilitate interaction and engagement with readers through comments and social media. A community can form around a blog, creating a space for discussions and shared interests.

6. Thought Leadership:

Professionals and experts in various fields use blogs to establish themselves as thought leaders. By sharing insights, research findings, and expert opinions, they can influence their industry and gain credibility.

7. Monetization:

Bloggers can monetize their blogs through various means, such as advertising, sponsored content, affiliate marketing, or even selling their own products or services.

8. Education and How-To Guides:

Blogs are often used to provide educational content and step-by-step guides. This could include tutorials, DIY projects, or in-depth explanations of complex topics.

9. Enumerate the steps to create your own blogs. (6)

=Creating your own blog involves several steps, from choosing a niche to selecting a platform and customizing your design.

Here's a step-by-step guide to help you start your own blog:

Step 1: Define Your Purpose and Niche

1. Choose a Topic (Niche):

Decide on the main theme or subject of your blog. Consider your passions, expertise, and the interests of your target audience.

2. Define Your Purpose:

Clarify the goals of your blog. Are you writing to inform, entertain, inspire, or sell products/services?

Step 2: Choose a Blogging Platform

3. Select a Blogging Platform:

Choose a platform that suits your needs. Popular options include WordPress, Blogger, Medium, and Wix.

4. Purchase a Domain Name:

Choose a unique and memorable domain name for your blog. You can purchase a domain through domain registrars like GoDaddy or Namecheap.

Step 3: Set Up Your Blog

5. Choose a Hosting Provider:

If you choose a self-hosted platform like WordPress.org, select a hosting provider. Bluehost and SiteGround are popular choices.

6. Install Your Blogging Platform:

Follow the instructions provided by your hosting provider to install your chosen blogging platform.

Step 4: Customize Your Blog

7. Select a Theme:

Choose a visually appealing and responsive theme for your blog. Most platforms offer a variety of free and premium themes.

8. Customize Design Elements:

Personalize your blog by customizing colors, fonts, and layout. Make sure your design aligns with your blog's theme and target audience.

Step 5: Create Compelling Content

9. Plan Your Content:

Develop a content plan. Decide on the types of posts you'll create, their frequency, and how they align with your blog's goals.

10. Write and Format Posts:

Create engaging and well-formatted blog posts. Use images, headers, and lists to make your content visually appealing.

Step 6: Optimize for SEO

11. Understand SEO Basics:

Learn the basics of Search Engine Optimization (SEO). Use keywords naturally in your content and optimize meta tags.

12. Submit Your Sitemap:

Submit your blog's sitemap to search engines like Google. This helps search engines index your content.

Step 7: Engage with Your Audience

13. Enable Comments:

Allow readers to leave comments on your blog posts. Respond to comments and engage with your audience.

14. Social Media Integration:

Share your blog posts on social media platforms. Integrate social sharing buttons on your blog for easy sharing.

Step 8: Monetize Your Blog (Optional)

1. Explore Monetization Options:

If you plan to make money from your blog, explore options like affiliate marketing, sponsored posts, or selling products/services.

2. Set Up Revenue Streams:

Implement revenue streams, such as Google AdSense for ads or affiliate marketing links, if applicable.

Step 9: Regularly Update and Promote

3. Consistent Posting:

Stick to a consistent posting schedule. Regularly update your blog with fresh and valuable content.

4. Promote Your Blog:

Actively promote your blog through various channels, including social media, email newsletters, and collaborations with other bloggers.